

BEDROCK ARCHITECTURE

Bedrock C ABI

C Language Binding and Calling Convention

Draft

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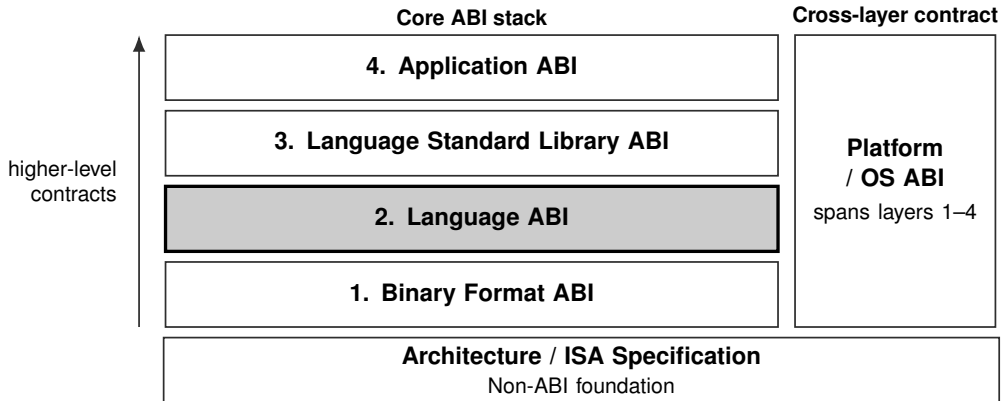
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1 SCOPE AND CONFORMANCE

This document defines the Bedrock C data model, register convention, calling convention, aggregate and variadic rules, compiler-runtime helper ABI, and C memory-model lowering. It depends on the Bedrock ELF ABI and the active platform/runtime environment ABI. Source syntax and semantics for target-specific C extensions are defined separately.

1.1 Contract Position



Current document: Bedrock C ABI. The shaded block identifies its primary contract position. The Platform / OS ABI spans and constrains the four core ABI layers; the ISA specification is their non-ABI architectural foundation.

1.2 Profile

Profile: `bedrock-c`

2 TERMINOLOGY

These terms are specific to the C binding. ELF and address-space terms keep the meanings defined by the Bedrock ELF ABI and the architecture manual.

object pointer	A 64-bit value that designates an object or one-past an object according to the C abstract machine. In the canonical interoperable context, the value is a pre-segment virtual address.
function pointer	A 64-bit value naming a C entry point. In the canonical interoperable context, the value is a pre-segment code address.
caller-saved register	A register whose value may be overwritten by a call unless the caller saves it before the call.
callee-saved register	A register whose incoming value must be preserved by the called function if the function uses it.
sret buffer	Caller-allocated storage used to receive a return value that is too large or unsuitable for the normal return registers.
addressable by-value argument	A by-value aggregate argument that is passed through caller-owned stack storage because the callee may need an address for the object.

3 DATA MODEL

Model:	LP64
Byte Order:	little endian
Internal SP:	8-byte aligned
Function Entry:	2-byte aligned
Before CALL:	$SP \bmod 16 = 8$
Aggregate Maximum Alignment:	16 bytes

3.1 Scalar Types

Table 3-1. Reference C Data Model

Type	Bits	Align
_Bool	8	1
char	8	1
signed char	8	1
unsigned char	8	1
short	16	2
int	32	4
long	64	8
long long	64	8
__int128	128	16
unsigned __int128	128	16
ordinary pointer	64	8
size_t	64	8
ptrdiff_t	64	8
wchar_t	32	4
char16_t	16	2
char32_t	32	4
float	32	4
double	64	8
long double	64	8
float _Complex	64	4
double _Complex	128	8
long double _Complex	128	8

3.2 Scalar Type Semantics

The widths and natural alignments of scalar types are given in Table 3-1. The following rules define the type identities and representations that are not apparent from those two values.

Integer types and aliases. Plain char is signed. size_t is an alias for unsigned long, and ptrdiff_t is an alias for long. wchar_t is a signed 32-bit integer; char16_t and char32_t are

unsigned 16-bit and 32-bit integers, respectively. In the absence of a separate enum extension, an enum has the representation, size, and alignment of `int`.

A stored `_Bool` has the integer value zero or one. False is encoded with every object-representation bit clear; true is encoded with bit 0 set and every other bit clear.

Ordinary pointers. An ordinary object or function pointer is 64 bits wide and 8-byte aligned. In the canonical interoperable context its value is a canonical pre-segment virtual address. Object and function pointers remain distinct C type categories even though their baseline representations have the same size and alignment. The null pointer has numeric address zero and an all-zero 64-bit object representation. Every ordinary-pointer result with numeric address zero is the null pointer.

128-bit integers. The types `__int128` and `unsigned __int128` are 16-byte, 16-byte-aligned integers. Their little-endian object representation places the low 64-bit word at the lower address. When the calling convention assigns one to a GENERAL-PAIR, the low word occupies the lower-numbered even register and the high word occupies the following odd register. A returned 128-bit integer therefore uses R0 for the low word and R1 for the high word. Its stack slot is 16-byte aligned.

An ordinary 128-bit load or store need not be atomic and may use two 64-bit memory operations. The 128-bit integer types are not supported as C atomic objects in the baseline profile.

Floating-point profile. The `bedrock-c` profile requires the base floating-point extension. Floating-point helpers may replace unavailable operations only when the F0–F15 register calling convention remains available.

Long double. `long double` is a distinct C type with the IEEE-754 binary64 object representation, 8-byte size, and 8-byte alignment. It uses the floating-point argument class and returns in F0, making its call representation ABI-compatible with `double`. In a variadic call it occupies one 16-byte slot containing an 8-byte binary64 value without widening.

Complex types. A complex object contains its real component first and its imaginary component second, each using the corresponding real type's representation. Therefore `float _Complex` has size 8 and alignment 4, while `double _Complex` and `long double _Complex` have size 16 and alignment 8.

3.3 Aggregate Layout

Aggregate layout applies the scalar and pointer sizes and alignments defined above as follows:

- Struct fields are laid out in declaration order with natural alignment up to the aggregate maximum.
- A union's alignment is the greatest alignment required by any member. Its size is the greatest ABI size of any member, rounded up to a multiple of the union alignment.
- Arrays store elements contiguously with each element using the ABI size of its type.

- Aggregate object size is rounded up to the aggregate object's alignment.

Bit-fields. A bit-field allocation unit has the size and alignment of its declared base type. Within the little-endian unit, allocation begins at the low bit. A field never crosses an allocation-unit boundary, and only consecutive fields whose base types are identical after typedef expansion may share a unit. A base-type change or a field that does not fit closes the current unit and begins a new naturally aligned unit. A field width may not exceed the representation width of its base type.

An externally visible bit-field base type shall be `_Bool`, `signed int`, `unsigned int`, or a typedef of one of those types. A plain `int` bit-field has the value range and representation of `signed int`. Other base types are target extensions and may not cross an external ABI boundary without a separately documented compatible extension ABI.

An unnamed nonzero-width field consumes space normally. A zero-width field closes the current unit and advances layout to the next boundary for its base type without adding storage of its own. Ordinary aggregate alignment, tail-padding, and size-rounding rules still apply after bit-field allocation.

Flexible arrays and excluded extensions. A flexible array member contributes no element storage to `sizeof` its containing structure. Its offset is aligned for the element type, and that element alignment participates in the structure's alignment calculation.

Packed aggregates and `#pragma pack` are not baseline ABI facilities. A compiler may expose them only as an extension, and such a type may not cross an external ABI boundary without a separately documented compatible extension ABI. Empty structures and zero-length arrays are likewise excluded from the baseline profile.

4 REGISTER CONVENTION

This section defines which architectural state survives an ordinary C call. Argument and result locations are defined by the assignment procedure in Section 5; they are intentionally not duplicated here.

4.1 Register Preservation

The caller shall preserve any live value held in a volatile register before a call. A callee that modifies a nonvolatile register shall restore its exact incoming value before returning. Registers not listed as nonvolatile carry no value across the call unless another rule in this ABI explicitly assigns the call result to them.

Table 4-1. C Call Preservation Sets

State	Volatile across a call	Preserved by the callee
General registers	R0–R7	R8–R15
Floating-point registers	F0–F7	F8–F15
Condition state	FLAGS	none
Segment state	GS1–GS5	CS, DS, SS, GS0

A callee may preserve floating-point registers with `FPUSHP` and `FPOPP` or with any sequence having the same observable effect. Each instruction operates on one canonical pair. The nonvolatile pairs are index 4 for `F8:F9`, index 5 for `F10:F11`, index 6 for `F12:F13`, and index 7 for `F14:F15`. When several pairs are saved, a normal prologue pushes them in increasing pair-index order and the matching epilogue pops them in decreasing pair-index order. A callee need only save the pair containing each nonvolatile register it modifies.

4.2 Stack and Control State

`SP` is the architectural stack pointer, not a general C temporary. A callee may move it while executing but shall restore the entry value before returning; Section 5 defines the additional alignment and frame rules. `PC` is the architectural program counter and is visible to C code only as a control-flow target. The call and return instructions, rather than a preservation class, define its transition across a call. Condition codes in `FLAGS` are volatile and may be clobbered by every callee.

4.3 Segment Context

An ordinary C call preserves the exact incoming images of `CS`, `DS`, `SS`, and `GS0`. The first three establish the code, ordinary-data, and stack contexts. `GS0` establishes the TLS context and may use translated-window mode. `GS1` through `GS5` are caller-saved scratch segment registers; no incoming value in them is preserved across a C call.

4.4 C Address-Space Contexts

The canonical interoperable C context uses disabled or bounds-only `CS`, `DS`, and `SS`. In this context, ordinary object and function pointers are canonical pre-segment virtual addresses and conforming execution maintains all of the following invariants:

- `CS`, `DS`, and `SS` are disabled with a zero mantissa or are enabled in bounds-only mode.
- Using an ordinary pointer as an effective address preserves its numeric value across segment processing.
- `DS` is disabled or its bounds contain every address that may be dereferenced through an ordinary object pointer, including materialized stack and TLS addresses.
- `CS` permits near-call reachability for every ordinary function entry in the active linkage domain, and `SS` covers the active stack.
- Every C object and its one-past address must be representable as a nonzero 64-bit address; no object may have mathematical one-past address 2^{64} .

The active platform ABI may instead execute C code with translated `CS`, `DS`, or `SS` images. That contract owns the pointer and symbol coordinate systems; compatibility among code, ordinary data, stack-derived addresses, and TLS-derived addresses; context transitions; and adaptation at every external boundary. In particular, independently translated `DS` and `SS` contexts do not by themselves make a stack-derived pointer usable as an ordinary data pointer. Code using such a context is not interoperable with the canonical context unless the active platform contract defines the necessary compatibility and transitions.

The 64-bit all-zero null-pointer representation remains unchanged in every context, and address zero is not assigned to a C object or function. Whether a null-derived access is guaranteed to fault is a mapping or guard-page policy of the active platform rather than a property of the representation.

4.5 Status Registers

STATUS is not part of the C abstract machine; ordinary C code accesses it only through target-specific intrinsics or runtime code. Low-level access to FSTATUS and FFLAGS uses `<bedrockfpintrin.h>`. FSTATUS is callee-saved: a function returns with the incoming rounding mode, floating-point exception-condition enables, and other control fields unless its documented interface changes the floating-point environment. FFLAGS is cumulative floating-point status and is caller-clobbered; floating-point operations in a callee may set accrued floating-point exception flags. An interface that requires a saved environment uses its floating-point environment runtime contract explicitly.

5 CALLING CONVENTION

This section is normative. A caller and callee conform when they independently apply the classification and assignment procedure below and therefore agree on every argument location, return location, and stack boundary. Register lists are summaries; they do not replace the assignment procedure.

5.1 Stack and Calls

The stack grows downward. An architectural CALL pushes an 8-byte return address at [SP], and callers align SP before the call so callee entry observes 16-byte alignment. Between ABI call boundaries, a function may adjust SP while keeping it aligned to at least 8 bytes. Immediately before CALL, the stack pointer satisfies $SP \bmod 16 = 8$ before return address push. After CALL pushes the return address, it satisfies $SP \bmod 16 = 0$ at callee entry.

Each object observes its declared type alignment. In particular, relaxing the internal stack-pointer invariant does not relax the 16-byte alignment of `__int128` objects or other objects whose declared alignment is 16 bytes. The ABI defines no red zone and no fixed outgoing-argument area. A callee that dynamically realigns the stack restores the incoming stack pointer before returning.

The first stack argument begins at [SP+16] at callee entry. The eight bytes at [SP+8] are alignment padding and carry no argument value. Each subsequent stack argument occupies the next 16-byte slot.

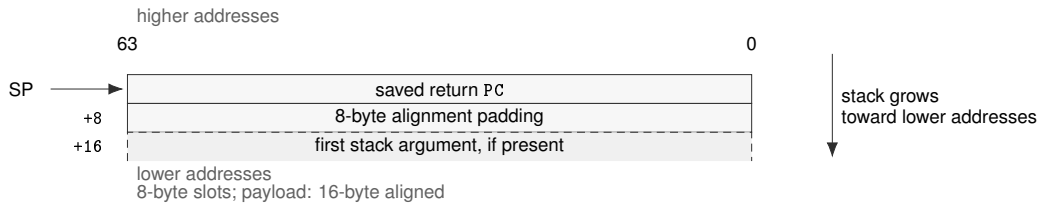


Figure 5-1. Near-call stack frame at callee entry

5.2 Frame Pointer

A function that maintains a frame pointer uses callee-saved R15. Frame pointers are otherwise omitted.

5.3 DWARF Register Numbering and Unwind

Table 5-1. Bedrock DWARF Register Numbers

Numbers	Registers	Status
0..15	R0..R15	assigned
16	SP	assigned
17	PC	assigned; CIE return-address register
18	FLAGS	assigned
19	STATUS	assigned
20..31	—	reserved
32..47	F0..F15	assigned
48	CS	assigned
49	DS	assigned
50	SS	assigned
51..56	GS0..GS5	assigned
57	FFLAGS	assigned
58	FSTATUS	assigned
59 and greater	—	reserved

At call entry, the canonical frame address is entry SP+8; saved PC is at CFA minus 8 and CS has a same-value rule. Signal and architectural event-frame unwind conventions belong to the active OS ABI.

5.4 Argument Classification

Classification occurs after the C language has determined the effective type of the argument. A named argument to a prototyped function is placed in one of five ABI classes:

GENERAL	Integer types no wider than 64 bits and ordinary object or function pointers. One general-register location is required.
GENERAL-PAIR	Signed and unsigned <code>__int128</code> . Two consecutive general registers beginning at an even-numbered register are required.
FLOAT	<code>float</code> , <code>double</code> , and <code>long double</code> . One floating-point register is required.
FLOAT-PAIR	<code>float _Complex</code> , <code>double _Complex</code> , and <code>long double _Complex</code> . Two consecutive floating-point registers are required, with no even-register alignment.
INDIRECT	Every structure or union argument, regardless of size or member composition. The caller creates a 16-byte-aligned by-value copy and passes the address of that copy as one GENERAL value.

Signed **GENERAL** values narrower than 64 bits are sign-extended when assigned to a register. Unsigned values and `_Bool` are zero-extended. Plain `char` follows its signed ABI type. A stack slot contains the effective type's object representation at its low address; bytes after that representation are unspecified.

5.5 Assignment Procedure

The caller shall apply the following procedure. The callee may rely on the result without inspecting any unused register or padding byte.

1. If the result uses an sret buffer, assign its address to `R0` and initialize the general-register cursor to 1. Otherwise initialize it to 0. Initialize the floating-point cursor to 0.
2. Visit arguments in source order. **GENERAL** and **INDIRECT** arguments consume the register named by the current general cursor and then advance that cursor. **FLOAT** arguments analogously consume `F0` through `F7` using the independent floating-point cursor.
3. For a **FLOAT-PAIR** argument, use the current floating-point register for the real component and the next register for the imaginary component, then advance the cursor by two. No even-register alignment is applied. If both registers are not available, place the complete complex value in one stack slot and mark the floating-point class exhausted.
4. For a **GENERAL-PAIR** argument, round the general cursor upward to an even register number. If that register and its successor both exist, place the low 64 bits in the even register, the high 64 bits in the odd register, and advance the cursor past the pair. A skipped odd register remains unused.
5. If an argument cannot be placed completely in the remaining registers of its class, place the complete argument in one stack slot and mark that register class exhausted. Later arguments of that class also use stack slots; they do not backfill a register hole. An argument is never split between registers and the stack.

6. Unnamed variadic arguments and every argument of an unprototyped call are forced to stack slots and do not consume register cursors. Apply the default argument promotions before assigning those slots.
7. Assign stack slots to stack-bound arguments in source order beginning at `[SP+16]`. Consequently, a caller that constructs the outgoing area by stores or pushes does so from the rightmost stack-bound argument toward the leftmost. No register home slots are reserved.

The general and floating-point resources exhaust independently.

5.6 C Return Values

Scalar results use their natural register class. A `__int128` result places its low word in `R0` and high word in `R1`. A structure or union of at most 16 bytes is returned as raw object bytes: increasing object addresses map first to `R0` and then to `R1`. Bytes beyond the object size and padding bytes whose values are not defined by C are unspecified.

A complex result places its real component in `F0` and its imaginary component in `F1`.

For a larger aggregate, the caller allocates a 16-byte-aligned result buffer and passes its address in `R0` before ordinary argument assignment. The callee stores the result in that buffer and returns the same address in `R0`.

Table 5-2. C Return Register Quick Reference

Result Class	Register Rule
General scalar	<code>R0</code>
Floating-point scalar	<code>F0</code>
long double scalar	<code>F0</code>
Complex scalar	real component in <code>F0</code> , imaginary component in <code>F1</code>
<code>__int128</code> scalar	<code>R1:R0</code>
Small aggregate	<code>R1:R0</code>
Large aggregate	sret pointer in <code>R0</code> , result pointer returned in <code>R0</code>

5.7 Aggregate Passing

For every by-value aggregate argument, the caller allocates distinct storage, aligns its start to 16 bytes, initializes it with the argument value, and keeps it alive until the call returns. The address, rather than the aggregate bytes, is assigned by the GENERAL procedure. If GENERAL registers are exhausted, the stack argument slot contains that address; it does not contain the aggregate itself.

The copy gives the callee an addressable object with the declared aggregate type. A callee may modify its private copy without modifying the caller's source object. Baseline aggregates containing bit-fields and mixed integer/floating aggregates use the same INDIRECT rule; their member composition never changes argument registers. An extension aggregate with packed or unaligned members cannot cross this external ABI boundary unless a separate extension ABI permits it.

5.8 Variadic Calls

Fixed named arguments use the ordinary assignment procedure. Every unnamed argument is first subject to the C default argument promotions and is then placed in a 16-byte stack slot. In particular, `float` becomes `double`, and `_Bool`, `char`, `short`, and their signed or unsigned variants promote to `int` when `int` can represent all values of the source type. `long double` remains the Bedrock binary64 `long double` type. An aggregate slot contains the address of the caller-owned by-value copy described above.

The baseline `va_list` representation is one 64-bit pointer to the next unnamed argument slot. `va_start` initializes it to the first unnamed slot after any stack-assigned named arguments. `va_arg` reads the promoted representation from the low bytes of the current slot, or follows the copy address for an aggregate, and then advances the pointer by 16 bytes. `va_copy` copies the pointer value and `va_end` has no machine effect.

An unprototyped call applies the same promotions and places every argument on the stack. As required by C, the callee type must be compatible with the promoted types.

6 FREESTANDING C BINDING

This section defines the compiler-runtime and ELF linkage obligations of the freestanding C profile. It does not define process startup or hosted-library behavior.

6.1 Execution Boundary

Process entry belongs to the platform/runtime environment ABI. BSS zero-fill, object loading, and relocation belong to the Bedrock ELF ABI. Hosted library behavior is outside the freestanding C ABI.

6.2 Compiler Runtime Helpers

The compiler lowers addition, subtraction, negation, bitwise operations, comparisons, truncation, and extension of 128-bit integers without an implicit runtime call. It may call one of the helpers below for 128-bit multiplication, division, remainder, and shifts; conversion between a 128-bit integer and a supported floating-point type; or complex multiplication and division.

Every compiler-runtime helper is an ordinary `bedrock-c` function. Its arguments and results therefore follow Section 5, including the `R1:R0` return convention for `__int128`.

The tables below are the complete baseline helper set. A conforming compiler shall not make an implicit reference to another compiler-rt or libgcc helper unless a separate target extension defines that symbol and its ABI. A runtime may export additional symbols, but their presence does not enlarge the baseline contract.

In the signature column, `i128` means `__int128`, `u128` means unsigned `__int128`, `c32` means `float _Complex`, and `c64` means `double _Complex`. The signatures use ordinary C parameter and result classification; a shift count has type `int`.

Table 6-1. __int128 Arithmetic and Shift Helpers

Operation	Helper	Signature
Multiply	<code>__multi3</code>	<code>i128(i128, i128)</code>
Signed divide	<code>__divti3</code>	<code>i128(i128, i128)</code>
Unsigned divide	<code>__udivti3</code>	<code>u128(u128, u128)</code>
Signed remainder	<code>__modti3</code>	<code>i128(i128, i128)</code>
Unsigned remainder	<code>__umodti3</code>	<code>u128(u128, u128)</code>
Shift left	<code>__ashlti3</code>	<code>i128(i128, int)</code>
Arithmetic shift right	<code>__ashrti3</code>	<code>i128(i128, int)</code>
Logical shift right	<code>__lshrti3</code>	<code>u128(u128, int)</code>

Table 6-2. __int128 and Floating-Point Conversion Helpers

Conversion	Helper	Signature
Signed to binary32	<code>__floattisf</code>	<code>float(i128)</code>
Signed to binary64	<code>__floattidf</code>	<code>double(i128)</code>
Unsigned to binary32	<code>__floatuntisf</code>	<code>float(u128)</code>
Unsigned to binary64	<code>__floatuntidf</code>	<code>double(u128)</code>
Binary32 to signed	<code>__fixsfti</code>	<code>i128(float)</code>
Binary64 to signed	<code>__fixdfti</code>	<code>i128(double)</code>
Binary32 to unsigned	<code>__fixunssfti</code>	<code>u128(float)</code>
Binary64 to unsigned	<code>__fixunsdfti</code>	<code>u128(double)</code>

Table 6-3. Complex Arithmetic Helpers

Operation	Helper	Signature
Binary32 multiply	<code>__mulsc3</code>	<code>c32(float, float, float, float)</code>
Binary32 divide	<code>__divsc3</code>	<code>c32(float, float, float, float)</code>
Binary64 multiply	<code>__muldc3</code>	<code>c64(double, double, double, double)</code>
Binary64 divide	<code>__divdc3</code>	<code>c64(double, double, double, double)</code>

The four scalar arguments to a complex helper are the real and imaginary parts of its two operands, in operand order. Bedrock long double has the binary64 representation and call classification, so conversions involving long double use the corresponding `df` helper and long double `_Complex` multiplication or division uses the corresponding `dc3` helper.

6.3 Pointers and External Objects

A C function pointer is 64 bits wide. In the canonical interoperable context, its value is a pre-segment near-call address. Code and data pointers have the same size and alignment, but remain distinct C type categories; neither representation creates an interchangeable code/data address space.

A common symbol has the ABI alignment of its declared type. An external object retains its declared alignment up to the 16-byte aggregate maximum. Stronger placement requirements belong to an explicit extension or platform contract.

6.4 Calls, Relocations, and Relaxation

The relocation records the linkage path selected for a C call:

Table 6-4. C Call Relocation Quick Reference

Call form	Relocation
Direct C call	R_BEDROCK_CALL32S
External call through the PLT	R_BEDROCK_PLT32S

The assembler and linker may relax an instruction sequence, but relaxation must preserve the C call interface, symbol identity, and observable control transfer defined by the original relocation. Relaxation is not a distinct C calling convention.

6.5 TLS and OS ABI Boundary

The Bedrock ELF ABI defines TLS models, symbols, and relocations. The active platform/runtime environment ABI defines thread setup, the initial GS0 image, loader handoff, and OS-facing TLS transitions.

7 C MEMORY MODEL

This section defines how the baseline C memory model maps to Bedrock memory operations and ordering instructions.

7.1 Ordinary Access Atomicity

The following guarantees apply to ordinary normal-memory accesses. They do not by themselves establish C inter-thread ordering:

Table 7-1. Ordinary Normal-Memory Access Guarantees

Access	Guarantee
Aligned 1/2/4/8-byte load or store	tear-free
Access of 16 bytes or more	not atomic
Unaligned access	no tear-free guarantee

7.2 Atomic Object Eligibility

The native lock-free set consists of integer objects and ordinary 64-bit pointer objects whose size is 1, 2, 4, or 8 bytes and whose address satisfies the natural alignment for that size. An unaligned atomic object is not lock-free. Accordingly, `__atomic_always_lock_free` returns true only for those naturally aligned sizes.

The baseline accepts an atomic object only when it belongs to that native lock-free set. Any other atomic object—including a floating-point, aggregate, 128-bit integer, unsupported-width, or insufficiently aligned object—is a target constraint violation.

7.3 Native Atomic Primitives

Once the object satisfies the width and alignment requirements above, the compiler uses the following primitive lowering. The ordering sequences are defined separately below.

Table 7-2. Native Atomic Primitive Quick Reference

C operation	Bedrock primitive
Atomic load or store	ordinary aligned tear-free access plus the required ordering sequence
Compare-exchange	width-matched <code>CMPXCHG</code>
Exchange	compare-exchange loop using <code>CMPXCHG</code>

7.4 C Atomic Memory Order Mapping

Table 7-3. C Atomic Memory Order Mapping

C order	Bedrock order or sequence
relaxed	relaxed
consume	acquire
acquire	acquire
release	release
acq_rel	acqrel
seq_cst	seqcst

For C compare-exchange, the compiler selects the join of the success and failure orders; it does not compare their numeric encodings. Consume is treated as acquire, acquire joined with release is acquire-release, and sequential consistency dominates every other order. A failure performs no write. Any release component introduced by the joined instruction therefore creates neither a write nor a release sequence on failure, while a failed `CMPXCHG.SEQCST` still participates in the global SC order as an SC load.

Table 7-4. Atomic Load and Store Ordering

Operation	Lowering
Relaxed load/store	no fence
Consume or acquire load	load, then <code>AFENCE</code>
Release store	<code>AFENCE</code> , then store
Sequentially consistent load	<code>AFENCE</code> , load, <code>AFENCE</code>
Sequentially consistent store	<code>AFENCE</code> , store, <code>AFENCE</code>

A relaxed C thread fence emits no instruction. Consume, acquire, release, acquire-release, and sequentially consistent thread fences emit `AFENCE`.

7.5 Volatile, Device, and Executable Memory

Volatile accesses preserve the observable access count and order but do not themselves provide inter-thread or device ordering. MMIO ordering belongs to platform/runtime mapping attributes. Dynamic code patching requires `__bedrock_sync_cache` or a platform interface with equivalent

cache synchronization; a relocation agent must synchronize patched instructions before transferring control to them.

7.6 C Fetch RMW Lowering

Table 7-5. C Fetch RMW Lowering

C operation	Lowering
<code>atomic_fetch_add</code>	<code>FETCHADD</code>
<code>atomic_fetch_sub</code>	<code>FETCHSUB</code>
<code>atomic_fetch_and</code>	<code>FETCHAND</code>
<code>atomic_fetch_or</code>	<code>FETCHOR</code>
<code>atomic_fetch_xor</code>	<code>FETCHXOR</code>

8 CALLING CONVENTION EXAMPLES

The examples illustrate the assignment procedure and do not introduce exceptions to the normative rules above.

8.1 Independent Register Classes and Pair Alignment

```
struct pair { uint64_t lo, hi; };
uint64_t mixed(uint64_t count, double scale,
    unsigned __int128 wide, struct pair pair, float bias);
```

The general cursor begins at 0 and the floating-point cursor begins at 0. `count` consumes R0; `scale` independently consumes F0. The GENERAL-PAIR argument rounds the general cursor from 1 to 2, so `wide` uses R3:R2 and R1 remains unused. The address of the caller-owned `pair` copy uses R4, and `bias` uses F1. No stack argument area is required.

8.2 sret Changes Ordinary Argument Assignment

```
struct triple { uint64_t x, y, z; };
struct triple build(uint64_t tag, signed __int128 wide,
    double factor);
```

Because the result is 24 bytes, its buffer address occupies R0 and the general cursor begins at 1. Thus `tag` uses R1, `wide` uses R3:R2, and `factor` uses F0. The callee stores the result through the entry value of R0 and returns the same address in R0.

8.3 A Pair That Does Not Fit Exhausts Its Class

```
void pressure(uint64_t a0, uint64_t a1, uint64_t a2,
    uint64_t a3, uint64_t a4, uint64_t a5, uint64_t a6,
    unsigned __int128 wide, uint64_t tail);
```

Arguments `a0` through `a6` use R0 through R6. The next even pair would begin past R7, so `wide` is placed completely at [SP+16] and the GENERAL class becomes exhausted. `tail` therefore uses [SP+32]; it does not backfill R7.

8.4 Variadic Promotions and Slot Traversal

```
int trace(const char *format, ...);
trace("example", (signed char)-1, (float)1.5, pair);
```

The named `format` argument uses R0. The signed character is promoted to `int` and stored at [SP+16]; the `float` is promoted to `double` and stored at [SP+32]. The caller creates the aggregate copy and stores its address at [SP+48]. A `va_list` initialized by `va_start` visits those three slots in that order by adding 16 after each access.

8.5 Assembly Conventions

The assembly examples are non-normative. File directives, basic-block comments, and local labels that do not affect the shown control flow are omitted. The scalar leaf example presents its one-instruction counted loop with `REP`. For a grouped body that does not write the selected loop-control register, REPG's post-body rule requires an equivalent form to bypass an initial zero but permits it to supply `N` directly for a positive `N`; the displayed `REP` also takes `N` directly and annuls zero. Apart from that zero-case normalization, instruction order and operands are unchanged. If an implementation sequence conflicts with a normative rule, the rule controls.

The declarations below state the ABI interfaces used by the examples.

8.6 Scalar Leaf Function

```
int sum(int *p, int n);
```

On entry, `p` is in `R0` and the sign-extended value of `n` is in `R1`. The function uses only volatile registers, does not change `SP`, and returns the integer sum in `R0`.

```
sum:
    mov.q  r0, r2
    clr.q  r0
    maxs.q r0, r1
    rep  r1, add.l [r2++], r0
    ret
```

Figure 8-1. Scalar leaf function

8.7 Non-Leaf Preservation and Call Alignment

```
int mf_pipeline(int *tmp, int *dst, int *src, int n,
               int ca, int cb, int cc);
```

The seven arguments occupy `R0` through `R6` in source order. The function keeps `n`, `dst`, `tmp`, and the first intermediate result live in nonvolatile `R8` through `R11`. It therefore saves and restores both canonical register pairs.

```
mf_pipeline:
    pushp    4
    pushp    5
    sub.q    8, sp
    mov.q    r3, r8
    mov.q    r1, r9
    mov.q    r0, r10
    mov.q    r2, r1
    mov.q    r8, r2
    mov.q    r4, r3
    mov.q    r5, r4
    mov.q    r6, r5
    call     mf_fir_three
    mov.q    r0, r11
    mov.q    r9, r0
    mov.q    r10, r1
    mov.q    r8, r2
    call     mf_scale_store
    add.l    r11, r0
    extzq.l  r0, r2
    mov.q    r9, r0
    mov.q    r8, r1
    add.q    8, sp
    popp     5
    popp     4
    jmp      mf_bias_sum
```

Figure 8-2. Non-leaf function with a tail transfer

Each PUSHBP saves 16 bytes, so the pair saves leave the entry alignment unchanged. The additional 8-byte subtraction establishes $SP \bmod 16 = 8$ before each CALL; the pushed return address therefore gives the callee a 16-byte-aligned entry stack. The epilogue removes the padding and restores R10:R11 before R8:R9. The final JMP is a legal tail transfer because the frame and non-volatile state have already been restored, the outgoing arguments have the ordinary ABI layout, and the target's R0 result is also the result required by mf_pipeline.

9 SYSTEM INTERFACE BOUNDARY

The C ABI does not define a hosted process-entry protocol or system-call ABI. Those contracts belong to the active OS ABI.

9.1 SYSCALL Boundary

Instruction: SYSCALL

Ordinary C code reaches SYSCALL only through OS ABI wrappers or target-specific runtime code.

9.2 OS ABI Segment Policy

Segment initialization and process policy are defined by the active OS ABI.